Complex Numbers

Section 1: Modulus-Argument Form

1 Find the polar representation of $z = -1 - i\sqrt{3}$.

a. $2(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3})$

b. $2(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3})$

c. $2(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3})$

d. $4(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3})$

2 What is the modulus of the quotient $\frac{1+3i}{2-i}$?

a. $\sqrt{2}$

b. 2

c. $\sqrt{5}$

d. $\frac{\sqrt{10}}{5}$

3 If $z = 4(\cos 70^\circ + i\sin 70^\circ)$ and $w = 2(\cos 40^\circ + i\sin 40^\circ)$, what is $\frac{z}{w}$?

a. $2(\cos 30^\circ + i\sin 30^\circ)$

b. $2(\cos 110^\circ + i\sin 110^\circ)$

c. $8(\cos 30^\circ + i\sin 30^\circ)$

d. $2(\cos 2800^\circ + i\sin 2800^\circ)$

Section 2: De Moivre's Powers

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4 Evaluate $(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}})^{10}$.

a. i

b. $-i$

c. 1

d. -1

5 What is $(\sqrt{3} - i)^6$?

a. -64

b. 64

c. $-64i$

d. $64i$

6 If $z + \frac{1}{z} = 2\cos\theta$, what is $z^n + \frac{1}{z^n}$?

a. $2\cos(n\theta)$

b. $2\sin(n\theta)$

c. $\cos(n\theta)$

d. $2^n\cos^n\theta$

Section 3: Roots of Complex Numbers

7 What is the principal square root of $8i$?

a. $2 + 2i$

b. $2 - 2i$

c. $4i$

d. $2\sqrt{2} + 2\sqrt{2}i$

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8 The four 4th roots of -1 form the vertices of what geometric polygon in the complex plane?

a. A square

b. A rectangle

c. A rhombus

d. A regular hexagon

9 If ω is a non-real cube root of 1, what is the value of $1 + \omega + \omega^2$?

a. 0

b. 1

c. -1

d. 3

Section 4: Complex Geometry & Loci

10 Describe the locus of points in the complex plane satisfying $|z - 3| = |z - 5i|$.

a. Perpendicular bisector of segment joining $(3, 0)$ and $(0, 5)$

b. Circle of radius 3

c. Ellipse with foci at 3 and $5i$

d. Ray starting at 3

11 What is the Cartesian equation for $|z - 2| = 3$?

a. $(x - 2)^2 + y^2 = 9$

b. $x^2 + (y - 2)^2 = 9$

c. $(x - 2)^2 + y^2 = 3$

d. $x^2 + y^2 = 9$

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12 Which inequality describes the interior of the unit disk in the complex plane?

a. $|z| < 1$

b. $|z| > 1$

c. $\mathrm{Re}(z) < 1$

d. $\mathrm{Im}(z) < 1$

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